

# Painter Naming Changes Feature Proximity and Variation: A Controlled Two-Painter Case Study

Anonymous authors  
Revised research manuscript prototype

7 September 2026

## Abstract

Painter-conditioned images may approach a reference collection while reproducing only part of its measured variation. We compare 1,006 generated images from two paid routes and an OAuth service with 70 digital painting surrogates by Monet and Cézanne, using 24 shared briefs, three repetitions and randomized condition orders. With 31 fixed interpretable features, all four paid-route painter-naming contrasts reduce empirical energy discrepancy and reject the prespecified conditional sharp null after Holm adjustment. A declared post-result revision then examines metric weighting, variation decomposition, reference influence, coverage and cross-route detection without adding inferential tests. Named/free total-trace ratios remain below one across four metric views and three processing pipelines, but within-brief contraction is not universal: OAuth Cézanne’s primary ratio is 1.139 despite aggregate contraction to 0.651. Removing texture reverses Nano Banana 2 Monet’s proximity improvement at 256 pixels and its below-reference spread at 512 pixels. Fixed-threshold cross-route balanced accuracy decreases under naming in every tested direction. All paid outputs are square and all references nonsquare, so absolute detectability is vulnerable to nuisance differences. These results support a finite-feature account of prompt-conditioned proximity and variation, without validating perceptual style collapse. The original tests and evidence are preserved; the revision generates zero images and spends no additional budget.

## 1 Introduction

The question motivating this study is whether painter-conditioned image generation reproduces the distribution of visual properties found across a painter’s original works. A small distance to a painter’s average feature vector does not answer this question: generated images can cluster tightly near that average while failing to reproduce variation across works. Conversely, a recognizable artistic reference may coexist with systematic differences in palette, spatial organization or digital texture.

We distinguish four properties: distribution discrepancy, within-set spread, reference coverage and painter specificity. None is an aesthetic quality score. The measured objects are digital reproductions of paintings and generated image files. Their comparison combines artistic choices with content, reproduction, resizing, encoding and service behavior. Treating every detectable difference as a failure of painter-style reproduction would overstate what the measurements identify.

Our experimental question is whether adding a painter’s name changes empirical feature discrepancy from that painter’s finite reference panel. Spread and detection address separate descriptive questions. We do not treat a broad claim of “style collapse” as an already validated hypothesis. The four paid-route contrasts provide conditional assignment evidence; subsequent diagnostics locate which parts of the observed contraction depend on measurement or comparison target. The original controlled study and the post-result revision are reported separately. All

review was maintainer-run LLM subagent review with coordinator checking, not independent human or institutional peer review.

## 2 Related work and the distinct research question

Quantitative art research has characterized painting collections using color, image statistics and information-theoretic measures (Kim et al., 2014; Lee et al., 2020; Sigaki et al., 2018). Kim et al.’s 2026 study concerns contextual multimodal navigation through art-historical development (Kim et al., 2026). Our sampling unit and target differ: we compare within-painter distributions across distinct works with newly generated text-to-image collections, with painter-name and prompt-content interventions. We do not claim to replicate its art-evolution analysis or its complete measurement system. Our fixed 31-feature representation is the existing implementation in this project.

Kim et al. analyze 72,447 paintings using learned formal/contextual vectors; their separate image-to-image experiment samples 500 source paintings per century and compares future-context tokens with an empty prompt. Our text-only painter-name intervention does not replicate that contextual instrument. AI-Pastiche evaluates artistic imitation with actual human authenticity and prompt-adherence surveys (Asperti et al., 2025a,b); detailed methods here follow its 12-generator, 73-prompt arXiv v1, with published metadata checked separately. AI-WikiArt instead uses 39,530 originals from 128 artists, captions of originals to condition three generators, and six VLMs for attribution judgments (Fu et al., 2025). VLM judgments are not an independent human panel. Our narrower contribution is the randomized prompt comparison and decomposition of empirical distribution changes, not greater artist breadth or human validation.

Asperti’s recent preprint reports separation of human and generated paintings in CLIP space and investigates preprocessing robustness, interpretable multiscale structure and inversion (Asperti, 2026). Thus, scatter separation alone is not sufficient novelty for this work. We emphasize painter-conditional comparisons, explicit artist-free controls and the effects of prompt content on a fixed interpretable representation. We have not run CLIP or reproduced that preprint’s inversion experiments. More generally, feature-space distinctions need not have the perceptual significance assumed by a style evaluator.

Generative-model evaluation separates fidelity and coverage for related reasons (Naeem et al., 2020). Resizing and compression can change measured differences (Parmar et al., 2022); digital-surrogate provenance is consequently part of the experimental design rather than a minor file-handling detail. Energy statistics provide a distributional comparison based on pairwise distances (Székely and Rizzo, 2013), without reducing each collection to its centroid.

### 2.1 Prior exposure and case selection

Earlier project analyses used 649 digital painting surrogates across Monet, Sisley, Pissarro and Cézanne and a 1,920-image prompt grid. Two late refusal retries completed its descriptive grid but did not restore its original complete-grid inference. Those exposed outcomes motivated additional model families and the present two-painter study. They are development evidence, not independent confirmation, and are not pooled into the new reference panel. The same 221 historical development works set the measurement scales. Painter selection and the revision choices do not support prevalence claims across artists or retrospective preregistration. The historical report retains the earlier results.

### 3 Controlled study design

#### 3.1 Choice of expansion axis

We expand model-family breadth and improve the comparison target before adding more painters. New collection focuses on Monet and Cézanne. The existing four painter results are retained as development evidence. This is a disclosed two-painter case study, chosen after earlier outcomes were known; it cannot estimate the prevalence of the phenomenon across all painters or traditions.

The research grid uses the same 24 content briefs across both painters, with three repetitions, named and artist-free conditions on three routes, plus a generic named condition on OAuth: 1,008 research slots. Eighteen separate technical requests and at most 24 bounded transient-error retries give a ceiling of 1,050 attempts, including at most 612 paid attempts. The spending ceiling is \$75, including the pilot and retries. Successful images are never regenerated or selected among alternatives. Sample counts reflect affordability and design coverage; they do not establish power for an image-space effect.

The reference target initially specified 48 fresh works per painter and 12 content strata. The recorded identity audit selects 64 Monet and 48 Cézanne candidates outside the prior exposure indexes. Original delivery encounters Wikimedia rate limits; a separately frozen successor follows its supported thumbnail policy. Nonupsampled standard delivery supports 38 Monet and 33 Cézanne candidates. All 71 deliveries succeed. Single-maintainer LLM visual coding excludes one figure-led Cézanne, leaving 38 Monet and 32 Cézanne works. These unequal counts are disclosed before measurement; no 48-work panel or fully balanced 12-stratum design is claimed.

Visually coded water, built and land organization replaces geographic-name heuristics. Counts are 21/4/13 for Monet and 3/11/18 for Cézanne. The primary comparison gives original works equal weight and generated images the same fixed class masses as their target reference. Eight shared briefs per class support this weighting. Collected-mixture and equal-class sensitivities are also required; the latter heavily weights Cézanne’s three water scenes. Mixed annotations remain flagged, with an exclusion sensitivity. This is coarse content control, not exact scene matching or independent expert annotation. Independent photographic capture pairs have not been established. Thumbnail and alternate-file provenance cannot substitute for evidence of distinct capture events.

#### 3.2 Generation routes, rendering and availability

The paid routes request Google `gemini-3.1-flash-image` and BFL `flux.2-max` through their pinned providers; the local OAuth route requests `gpt-image-2`. Six separate technical pilot calls per route verify transport before research collection. Requested identifiers do not attest immutable weights.

The controlled collection retains 1,006 images from 1,008 fixed slots: 288 Nano Banana 2, 288 FLUX.2 Max and 430 OAuth outputs. Two OAuth generic-named Cézanne slots receive moderation refusals and remain missing. One complete FLUX 502 receives one successful exact-payload technical retry under the fixed rule. Including 18 pilot attempts, there are 1,027 physical attempts, 589 paid. All outcomes, costs and retries remain in the retained transport receipts. No successful output is regenerated or selected among visual alternatives.

All paid research outputs are 1024-square: Google JPEG and BFL PNG. OAuth returns 430 PNGs with short sides 1087–1254 pixels; eight are square. Of these, 424 report low quality and six report medium despite all requesting medium. The six medium outputs are artist-free (three per painter); all named outputs report low. These response fields do not attest an immutable model or comparable rendering. OAuth is therefore a separate service comparison. A prompt

contrast on that route includes any prompt-induced service handling; it cannot isolate painter naming at fixed delivered quality. No output is filtered by its reported quality. Symmetric 256-pixel and common-JPEG sensitivities probe processing effects but cannot undo earlier encoding or generation-quality differences.

## 4 Distribution measurements and statistical analysis

### 4.1 Feature representation and processing

Each image is represented by 11 color, eight spatial and 12 digital-texture features. Color includes lightness and chroma location/spread, hue organization, and multiscale chromatic differences. Spatial features include spectral shape, orientation organization, gradients and quadrant differences. Texture includes wavelet energies and trends, local binary-pattern entropies and local luminance variation. These are measurements of digital images, not direct measurements of physical brushstroke relief or a validated complete definition of painter style.

The primary normalization applies EXIF orientation, converts valid embedded profiles to sRGB, assumes sRGB when compatible unprofiled data are present, rejects nonopaque images and preserves aspect ratio at a 512-pixel short side. It uses Lanczos resampling without upsampling or discretionary cropping. Each coordinate is centered by the development median and divided by the development interquartile range. Development painters receive equal total mass; no generated or new reference feature is used to fit that scaler.

### 4.2 Distribution discrepancy and spread

Let  $x_i$  and  $y_j$  be standardized reference and generated vectors, with nonnegative weights  $a_i$  and  $b_j$  that each sum to one. We report the empirical V-statistic energy discrepancy

$$\begin{aligned} \widehat{\mathcal{E}}(X, Y) = & 2 \sum_{i,j} a_i b_j \|x_i - y_j\|_2 - \sum_{i,k} a_i a_k \|x_i - x_k\|_2 \\ & - \sum_{j,l} b_j b_l \|y_j - y_l\|_2. \end{aligned} \tag{1}$$

All pairwise terms contribute; the reference mean is not the sole comparison target. The finite-sample V-statistic has a nonzero sampling baseline even when independent samples come from one common population.

We report the ratio of total within-group sample variances for ordinary unweighted data. Weighted analyses use the population-form trace  $\sum_i a_i \|x_i - \sum_k a_k x_k\|^2$  in both numerator and denominator. The sample and population definitions have different finite-sample denominators. A complementary robust summary divides the sum of squared generated coordinate IQRs by the corresponding reference sum. Undefined coordinate ratios remain undefined rather than being stabilized by an arbitrary constant.

### 4.3 Content control and sensitivity analyses

The controlled reference panel has now produced all 210 expected feature vectors under three processing pipelines, without measurement failures or duplicate raw image hashes. Recomputing those vectors from retained source bytes reproduces all records. Both sensitivity scalars were fitted first, from 442 remeasurements of the same 221 historical development works. All 1,006 selected generated images produce their 3,018 expected vectors without measurement failures or duplicate raw hashes. Generated features are first accessed after terminal collection, and retained raw bytes support exact replay.

For painter  $p$ , let  $N_{pc}$  denote the number of reference works in content class  $c$ ,  $N_p$  the total, and  $n_{pc}$  the number of available generated observations in a particular route/condition cell. The primary weights are

$$a_i = 1/N_p, \quad b_j = \frac{N_{pc}}{N_p n_{pc}} \quad \text{for } j \in c. \quad (2)$$

Each cell is assigned 72 slots: eight briefs per content class, each repeated three times. Missing outputs redistribute only their own class’s fixed mass; a required class with no measured output makes that comparison unavailable. The weighting is conditional on intended generated content, whose adherence has not been independently judged. Images are not discarded for failing to resemble a painter or for depicting the intended scene imperfectly. Effective weight size  $1/\sum_j b_j^2$  is reported separately from the number of image files.

Primary energy discrepancy and the two spread ratios use all 31 standardized features. Color, spatial and texture subsets remain descriptive. Prespecified sensitivities use the collected generated mixture, one-third mass per content class, only unambiguous reference annotations, only retained reference files with recorded parent-surrogate short sides at least 1024 pixels, and initial-attempt successes alone. The annotation and native-size restrictions redefine their reference class masses from the retained subset and are labeled accordingly. A 256-pixel short side and common JPEG encoding at quality 90 with 4:2:0 subsampling additionally probe processing sensitivity. No histogram, palette or texture matching is performed.

For the new disjoint original/original baseline, each of 999 deterministic draws selects two nonoverlapping sets of  $\lfloor N_{pc}/2 \rfloor$  works per class and a generated set with the same class counts, without replacement. All sets retain the declared class masses. The resulting group sizes are 18 for Monet and 15 for Cézanne; the latter includes only one water work per group. The repeated draws characterize this finite panel and do not create independent reference samples.

Secondary coverage uses an original-centered ball with radius  $r_i$  equal to the distance to the third nearest distinct reference work:

$$\hat{C} = \sum_i a_i \mathbf{1} \left\{ \min_j \|x_i - y_j\|_2 \leq r_i \right\}. \quad (3)$$

We report all-available coverage and 100 uniform generated subsamples of size  $\min(N_p, n_p)$ , without replacement. These neighborhood measurements depend on sample size and representation and are not artistic-completeness scores.

Fixed linear/RBF kernel-ridge detection uses six whole-brief folds and disjoint reference works, with no tuning or PCA input. Reference folds are assigned by content-stratified identity hashes, so sparse classes are distributed across folds. Training assigns equal total mass to original and generated labels and matches content masses within each label; held-out balanced accuracy and AUC use the same target masses. Eight folds over assigned collection batches provide a timing sensitivity; the batches do not identify independent service sessions. Every split and prediction is retained. For visualization, each painter has one common PCA fit with half the mass on originals and half shared equally across seven generated cells, using the primary content weights. Original-only PCA is also retained. Every point is shown under shared within-painter axes. Painter specificity compares each named cell with both reference painters under the same one-third-per-class mixture, separately from its absolute discrepancy.

#### 4.4 Assignment, execution and missingness

The reference, prompt inventory, endpoints, scaling and analysis are committed before evaluation features are extracted. Repetitions are assigned to eight batches with originally planned offsets 0, 3, 6, 9, 24, 27, 30 and 33 hours. Within-route condition order is randomized, with at most

one active call per route and globally staggered starts at least five seconds apart. All attempts, refusals, technical retries and actual timestamps are preserved.

At the user’s request, the remaining four batches ran consecutively rather than waiting for the originally planned offsets. Original assignments and actual timestamps remain evidence; the full 33-hour schedule was not executed. Closed collection components and their separately bound corrections are retained, with all decisions preceding generated-feature measurement. Assigned batches do not identify independent backend sessions. The revision examines timing sensitivities without changing the original randomization calculation.

Prospective synthetic qualification passes for the eight paired all-31 prompt contrasts: six named-versus-free and two detailed-versus-generic OAuth contrasts. Two fixed-seed phases each use 10,000 trials in 24 null cells covering pair counts, contribution heterogeneity, window drift and endpoint dependence. Actual tests use 99,999 Monte Carlo sign draws and Holm adjustment. Their sharp null concerns availability and features under the randomized slot policy, conditional on the finite reference, with no interference. Service carryover and retry-induced timing remain limitations. This qualification does not validate absolute-distribution p-values, oeuvre-sampling intervals or actual image-space power. Absolute gaps and leave-batch/brief ranges remain descriptive. Ordinary image-level bootstrap intervals are not justified by the number of files alone. No significance-based sampling extension, silent replacement of missing outputs or retrospective change to a sealed endpoint is allowed. Human construct validation remains unperformed. Absolute discrepancy and spread are descriptive; only the eight prespecified prompt contrasts carry conditional randomization p-values.

## 4.5 Paired statistic and conditional interpretation

For a matched before/after pair  $(A_i, B_i)$  with fixed class-adjusted weight  $b_i$ , define  $C(v) = \sum_j a_j \|v - x_j\|_2$  and  $T(v) = \sum_i b_i (\|v - A_i\|_2 + \|v - B_i\|_2)$ . The contribution

$$c_i = b_i \{2[C(B_i) - C(A_i)] - [T(B_i) - T(A_i)]\} \quad (4)$$

satisfies  $\sum_i c_i = \hat{\mathcal{E}}(X, B) - \hat{\mathcal{E}}(X, A)$ . Thus the contrast retains the within-generated pairwise terms; it is not a contrast of mean-to-centroid distances. Complementary condition swaps produce  $\sum_i s_i c_i$  for assignment signs  $s_i \in \{-1, 1\}$ . For the three-condition OAuth groups, each pairwise comparison conditions on the third condition’s position in the uniformly randomized order.

A contrast uses complete measured pairs sharing a brief, repetition and assigned batch. If fewer than 24 pairs or 12 distinct briefs survive, or a required content class is absent, its endpoint is unavailable and retains  $p = 1$  in the fixed eight-test family. This is distinct from the all-available descriptive comparison. Two-sided Monte Carlo p-values use  $(1 + e)/(1 + 99999)$ , where  $e$  counts sign-draw statistics at least as extreme in absolute value, including numerical ties. Leave-one-batch and leave-one-brief estimates assess sensitivity without being labeled confidence intervals. The sharp assignment null covers both availability and measured features for the full fixed slot policy, including its bounded technical retry rule; it is stronger than equality of expected distances. Treatment-dependent service duration and retry delays can affect later calls, so the randomization calculation does not remove the no-interference limitation.

## 5 Controlled results

### 5.1 Proximity and spread are distinct

All 14 primary cells and all eight inferential endpoints are available. Table 1 compares the full distributions, not just painter means. Every named condition has less total empirical variance

than its reference (ratios .306–.756). Detailed named ratios are .354–.756; the two generic-named OAuth cells are .306 and .395. Naming also contracts spread relative to the corresponding artist-free condition in all six route/painter comparisons. However, paid artist-free ratios are 1.069–1.535. The data therefore support concentration of these painter-conditioned collections, not an intrinsic claim that generated images always have less variation.

| Painter | Route | Prompt  | Energy | Variance | IQR sum | RBF BA |
|---------|-------|---------|--------|----------|---------|--------|
| Monet   | NB2   | Free    | 4.188  | 1.086    | 1.131   | 0.939  |
|         | NB2   | Named   | 3.342  | 0.756    | 0.966   | 0.906  |
|         | FLUX  | Free    | 1.990  | 1.071    | 0.941   | 0.901  |
|         | FLUX  | Named   | 1.365  | 0.354    | 0.377   | 0.953  |
|         | OAuth | Free    | 2.567  | 0.863    | 1.176   | 0.960  |
|         | OAuth | Named   | 2.333  | 0.478    | 0.586   | 0.962  |
|         | OAuth | Generic | 2.370  | 0.306    | 0.330   | 0.987  |
| Cézanne | NB2   | Free    | 4.704  | 1.535    | 1.371   | 0.970  |
|         | NB2   | Named   | 2.886  | 0.716    | 0.785   | 0.977  |
|         | FLUX  | Free    | 1.700  | 1.069    | 0.950   | 0.903  |
|         | FLUX  | Named   | 0.804  | 0.492    | 0.567   | 0.893  |
|         | OAuth | Free    | 1.941  | 0.787    | 0.928   | 0.958  |
|         | OAuth | Named   | 1.884  | 0.512    | 0.461   | 0.920  |
|         | OAuth | Generic | 2.166  | 0.395    | 0.299   | 0.918  |

Table 1: Primary 512-pixel, all-31, reference-content-weighted results. Variance and squared-IQR-sum values are generated/original ratios. BA holds out whole briefs and distinct original works. Each cell has 72 images except Cézanne/OAuth/generic (70); references have 38 Monet and 32 Cézanne works. NB2 denotes Nano Banana 2; FLUX denotes FLUX.2 Max; OAuth is the GPT Image 2 service. Named uses detailed briefs; Generic uses generic named briefs.

The contraction is uneven across feature families. Among the eight named cells, color variance ratios range .291–.745 and digital texture .143–.460. Spatial ratios range .665–1.520, with four detailed named cells above one. Thus, lower aggregate spread does not mean lower variation in every type of visual property. These family results are descriptive and do not identify a physical brushstroke or perceptual-diversity mechanism.

## 5.2 Scatter overlap coexists with full-space detectability

The balanced joint PCA retains 48.4% of weighted variation in its first two components for Monet and 46.0% for Cézanne. The clouds overlap in these views (Figure 2), with route- and condition-specific shifts and concentration. Original-only PCA, retained in the report, explains 55.6% and 52.2% in two dimensions and also shows overlap. Neither projection supports a claim of perfectly disjoint distributions.

In all 31 features, whole-brief/work-disjoint RBF balanced accuracy is .893–.987 across all 14 cells, including artist-free controls. Linear accuracy is .837–.969. Holding out assigned batches gives .889–.987 (RBF) and .822–.969 (linear). These fixed-model descriptive scores establish practical detectability within the recorded evaluation splits. They do not identify which differences represent painter style, nor provide an absolute distribution-equality test.

## 5.3 Painter naming improves primary 31-feature proximity on the paid routes

All six primary named-minus-free energy point estimates are negative (Table 2). The four paid-route contrasts reject the fixed conditional sharp null after Holm adjustment over eight tests. Their improvements range from .625 to 1.818 energy units. Every leave-one-brief and leave-one-

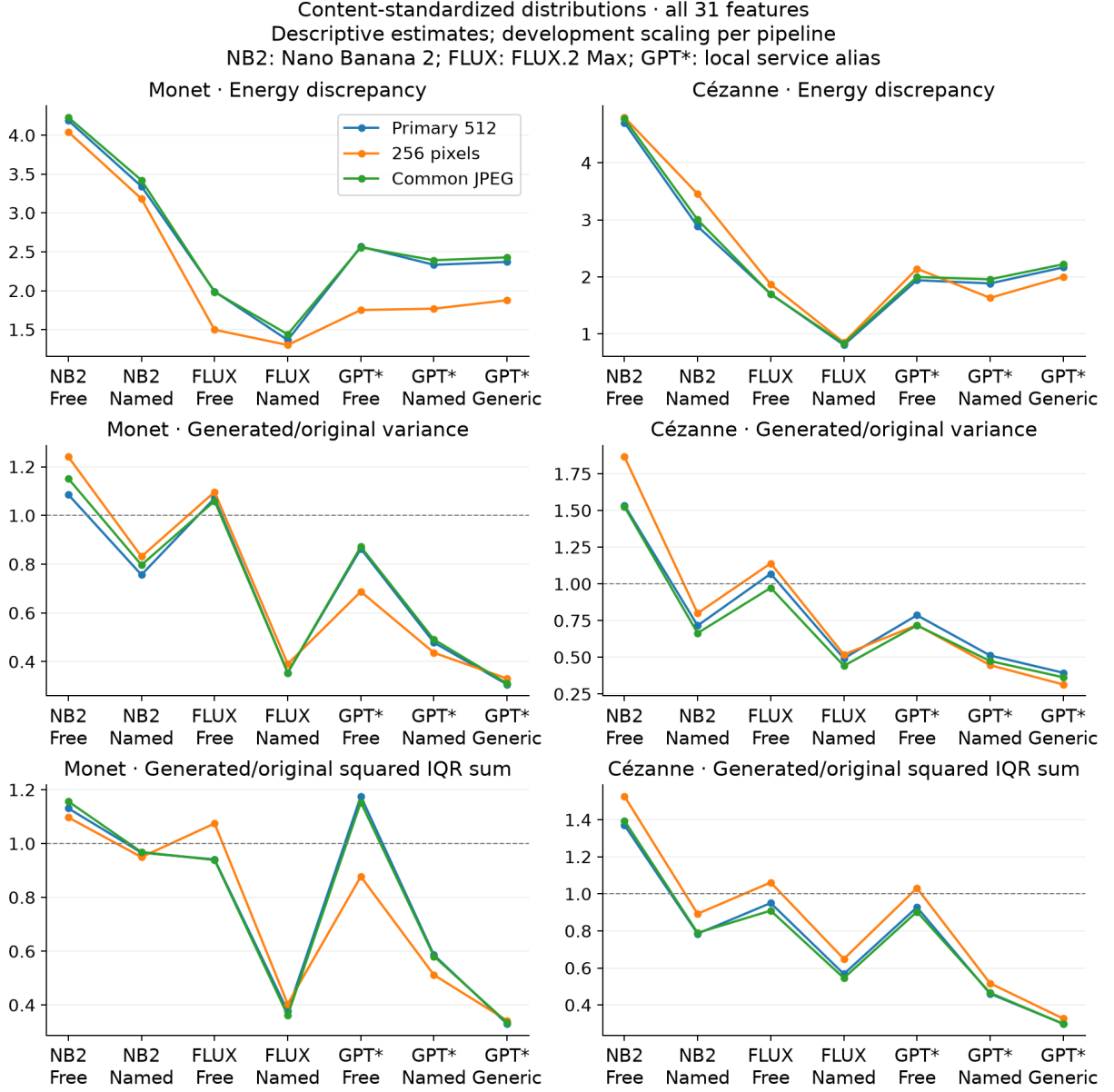


Figure 1: Full-distribution discrepancy and two spread summaries across the three fixed processing pipelines. Named means detailed named; Generic means generic named. Every pipeline uses its own development-only scaler. Dashed lines mark a spread ratio of one, not a calibrated equivalence boundary.



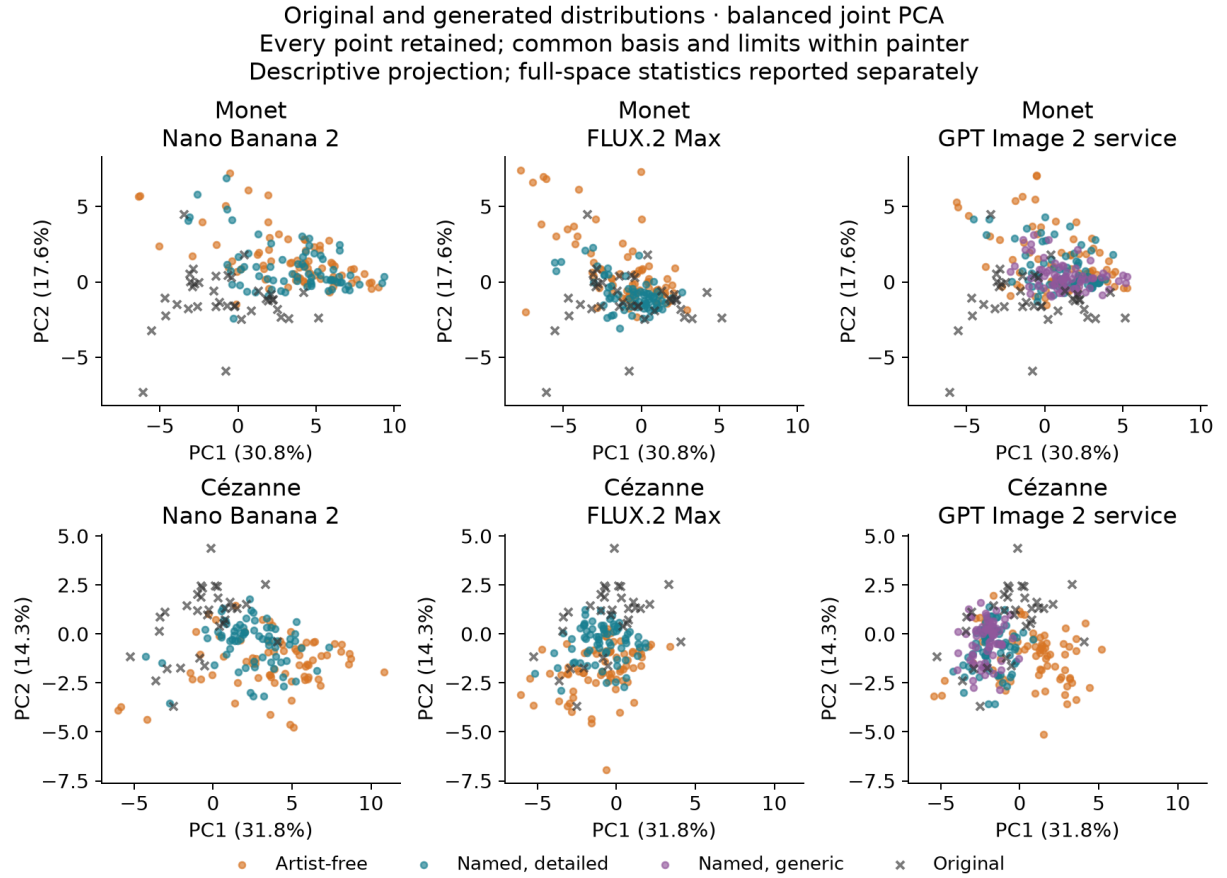


Figure 2: Common painter-specific PCA with every original and generated point retained. Each painter uses one basis and shared limits across routes. Originals receive half the fit weight; generated cells share the other half. Overlap in two projected dimensions coexists with strong held-out full-space detection.

batch estimate for those four contrasts remains negative. These ranges describe the finite design; they are not confidence intervals.

Neither OAuth naming contrast rejects after adjustment (Holm  $p = .19308$  for Monet and  $p = 1$  for Cézanne). The generic-to-detailed OAuth contrast is small for Monet ( $-.0375$ , Holm  $p = 1$ ) and larger for Cézanne ( $-.2723$ , Holm  $p = .0928$ ), using 70 complete Cézanne pairs because both refusals occur in its generic condition. Non-rejection does not establish no effect. Detailed named OAuth images have more empirical spread than generic ones for both painters, but this comparison changes content specificity as well as wording; it is not a pure length effect.

| Route | Painter | Contrast | Pairs | Energy change | Raw $p$   | Holm $p$           |
|-------|---------|----------|-------|---------------|-----------|--------------------|
| NB2   | Monet   | Name     | 72    | -0.8462       | 0.00062   | 0.0031             |
| NB2   | Cézanne | Name     | 72    | -1.8176       | $10^{-5}$ | $8 \times 10^{-5}$ |
| FLUX  | Monet   | Name     | 72    | -0.6246       | $10^{-5}$ | $8 \times 10^{-5}$ |
| FLUX  | Cézanne | Name     | 72    | -0.8956       | $10^{-5}$ | $8 \times 10^{-5}$ |
| OAuth | Monet   | Name     | 72    | -0.2339       | 0.06436   | 0.19308            |
| OAuth | Cézanne | Name     | 72    | -0.0561       | 0.61624   | 1                  |
| OAuth | Monet   | Detail   | 72    | -0.0375       | 0.773     | 1                  |
| OAuth | Cézanne | Detail   | 70    | -0.2723       | 0.0232    | 0.0928             |

Table 2: Prespecified two-sided paired tests, 99,999 Monte Carlo sign draws, Holm family size eight. Name compares artist-free with detailed named; Detail compares generic named with detailed named. Negative values mean closer after the intervention, on complete matched pairs. The sharp null includes availability and features under the fixed slot/retry policy and assumes no interference.

## 5.4 Finite-reference baselines, coverage and specificity

Across the 14 matched-sample comparisons, original/original energy medians range .697–.753, while original/generated medians range 1.170–5.232. Every generated comparison has a larger median than its corresponding original split. The lowest generated median is Cézanne/FLUX/named (1.170 versus .718); its subsample ranges overlap the original baseline. These finite-panel draws do not establish a population rejection or an equivalence threshold.

FLUX named outputs have the lowest primary energy among the tested routes for both painters (1.365 Monet, .804 Cézanne), together with the highest named all-available neighborhood coverage (.789 and .750). Equal-size subsample coverage medians are lower (.737 and .672). For Monet/FLUX, naming improves energy from 1.990 to 1.365 and coverage from .553 to .789, while variance contracts from 1.071 to .354. Proximity, spread and neighborhood coverage can therefore move in different ways; a single distance cannot describe all three.

Under equal content-class masses, five of six detailed named cells are closer to their own reference painter than to the other painter. Nano Banana 2’s Monet condition is the exception: energy 3.687 to Monet versus 3.488 to Cézanne. Improved proximity to a requested target thus does not guarantee painter specificity. This two-painter comparison is descriptive and is not a validated recognition task or a general model leaderboard. This original own/other ranking includes reference-specific within-distribution terms. The post-result double contrast below cancels those terms; its negative interaction and this individual nearest-reference exception answer different questions.

## 5.5 Processing and reference sensitivities

For the eight named cells, total-variance ratios remain below one under both fixed processing sensitivities: .315–.831 at 256 pixels and .310–.797 after common JPEG encoding. All four

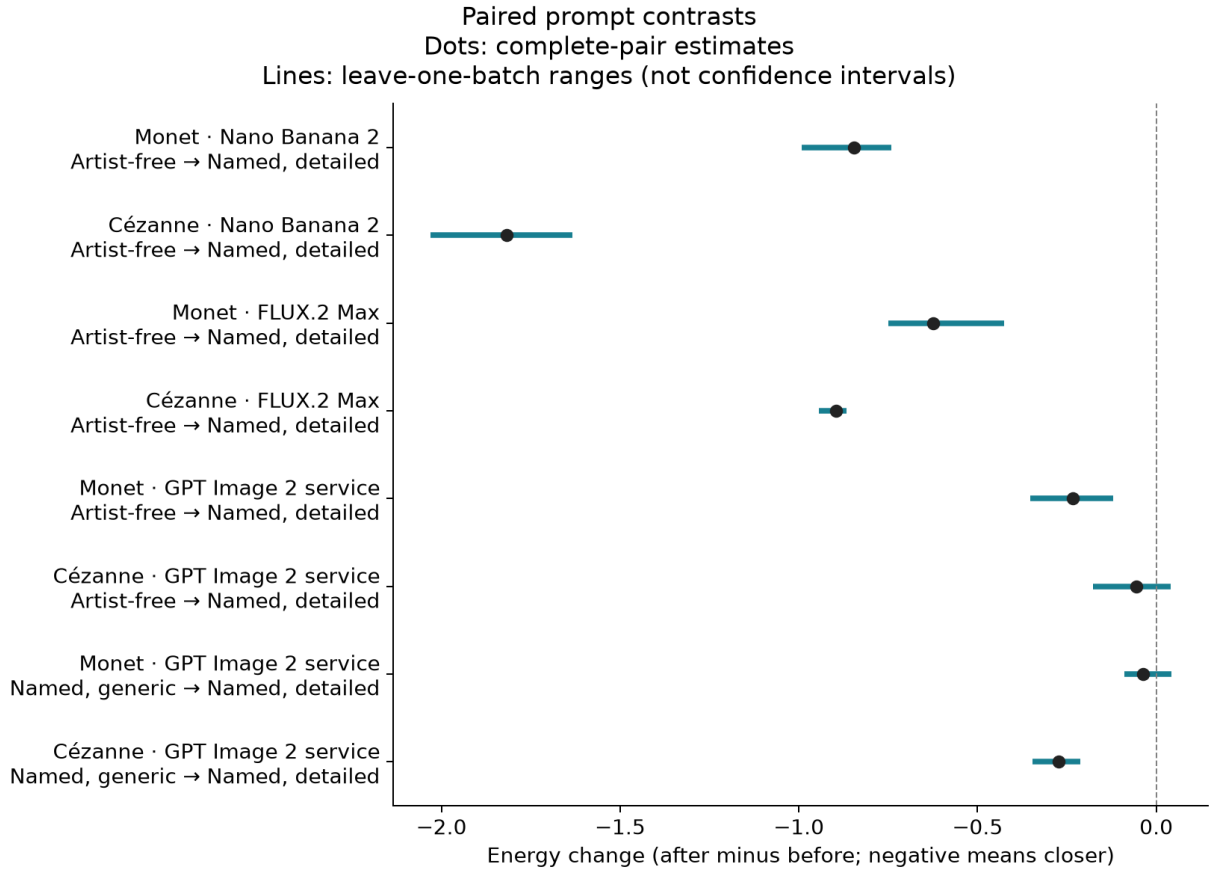


Figure 3: Complete-pair estimates and leave-one-assigned-batch ranges. Shared service state and timing dependence remain limitations of the conditional tests; the displayed lines are not uncertainty intervals for a painter’s oeuvre.

paid-route naming effects retain their negative descriptive energy direction under both. OAuth is less stable: Monet’s naming difference changes from  $-.234$  in the primary pipeline to  $+.017$  at 256 pixels. No additional significance family is introduced for these sensitivities.

Collected-mixture, equal-content and unambiguous-reference analyses retain aggregate variance ratios below one for all eight named cells. Some IQR-based ratios exceed one after reweighting or reference restriction. The native-1024 restriction is particularly limited: only four Monet and 19 Cézanne reference files remain; the four Monet works contain two water, no built and two land scenes. These counts use parent metadata, not delivered thumbnail dimensions, which would retain 18 Cézanne files. This changes content support as well as resolution metadata. Nano Banana 2’s Monet variance ratio becomes 1.107 in that primary subset (1.123 at 256 pixels; 1.178 after JPEG). This is a material exception to uniform robustness and a change in the finite comparison target. It cannot serve as a strong independent capture control. Initial-only analysis preserves all named cells; its only additional exclusion is the retried FLUX artist-free slot.

## 6 Post-result methodological revision

### 6.1 Declared scope and measurement alternatives

After the controlled results and skeptical review, a separate namespace fixed the diagnostic methods, source, tests, input hashes and seed before running the full revision. These choices were informed by exposed results: this is not a retrospective preregistration. Every new result is descriptive, with no additional confirmatory tests, significance claims or replacement of the original eight-test family. The calculations read retained vectors and metadata, without opening pixels, extracting new features or making provider requests.

Four fixed feature views are applied to all three processing pipelines: original 31 coordinates; 28 coordinates omitting three derived summaries (deltaE slope, wavelet slope and curvature); equal coordinate-count family weighting, dividing color/spatial/texture coordinates by  $\sqrt{11}$ ,  $\sqrt{8}$  and  $\sqrt{12}$ ; and 19 color/spatial coordinates without texture. Family weighting does not equalize observed variance or validate style relevance. All 168 cell summaries and 72 named/free contrasts are retained. Energy magnitudes across differently scaled views are not directly comparable.

For generated vector  $z_{cbr}$ , class  $c$ , brief  $b$  and repeat  $r$ , let  $w_c$  be its class mass,  $u_{b|c}$  its conditional brief mass and  $v_{r|cb}$  its conditional repeat mass. With corresponding means, total trace decomposes as

$$T_G = \sum_{cbr} w_c u_{b|c} v_{r|cb} \|z_{cbr} - \mu_{cb}\|^2 + \sum_{cb} w_c u_{b|c} \|\mu_{cb} - \mu_c\|^2 + \sum_c w_c \|\mu_c - \mu\|^2. \quad (5)$$

These are within-brief, between-brief within-class, and between-class components. Between-brief-means trace combines the last two. Reference trace has only within/between-class components; original brief labels are not invented. These are exact weighted finite-set identities, not unbiased variance-component estimates from three repeats. Coordinate/family trace contributions, development correlations and four development-painter-omission scalars accompany the results. Alternative scales use development data only.

Primary original-31 influence deletes each reference work, each exact set of recorded holding-collection IDs, or each content class. Work/source deletions keep the original class masses where support permits; class deletions renormalize the remaining target. An exact collection-membership set is a source proxy, not a documented photographic workflow. Deletion ranges are not confidence intervals.

Under common one-third class masses, the painter-prompt/reference double contrast is

$$I = \mathcal{E}(X_A, G_A) + \mathcal{E}(X_B, G_B) - \mathcal{E}(X_A, G_B) - \mathcal{E}(X_B, G_A). \quad (6)$$

Reference-only and generated-only within terms cancel. Negative  $I$  indicates relative own-painter alignment in these features, not human recognition. We also calculate  $I$  for the separately collected artist-free groups and their pairwise energy, checking exact canonical payload identity for all 72 matched slots per route. This collection-label placebo is not a calibrated null or an equivalence threshold.

Coverage uses  $k = 1, 3, 5$  and 100 fixed-seed draws, half/full reference sample sizes and uniform/class-count-matched generated sampling. Per-anchor radii, same/cross-class hits and all memberships are retained. A separate control splits originals into disjoint anchor/query sets with  $\lfloor N_{pc}/2 \rfloor$  per class and matches generated query counts to those anchors. This defines a different neighborhood target from full-panel coverage; its 18/15-work groups are not independent replications across draws.

A square metadata rule and within-stratum feature/descriptor associations assess nuisance plausibility. No causal adjustment is claimed without common support. Directed cross-route detection uses the inherited fixed linear/RBF kernels and ridge, six disjoint original-work/whole-brief folds, and all 24 shared-condition route/painter cases. We report pooled fixed-threshold balanced accuracy and per-fold AUC, never a pooled AUC across differently fitted margins. Verified capture-workflow holdout is explicitly unavailable. All eight original endpoints also receive descriptive initial-only, component-boundary, over-120-second group-span, and combined boundary/retry exclusions. OAuth spans include all three randomized conditions. These exclusions do not repair interference. The fixed seed is 2026090719, with stable named substreams and no seed search.

## 6.2 Aggregate contraction does not imply within-brief contraction

Table 3 locates the primary named/free change. All six total trace ratios are below one, but OAuth Cézanne has a within-brief ratio of 1.139: its reduced aggregate spread is driven by differences between brief means. Nano Banana 2 Cézanne contracts between-brief means much more than within-brief variation. FLUX contracts both components strongly for both painters. Thus the results support aggregate concentration, but not uniformly reduced stochastic variation at a fixed prompt. Three repetitions give limited information about each brief’s generating distribution.

| Route | Painter | Total trace | Within brief | Between brief means |
|-------|---------|-------------|--------------|---------------------|
| FLUX  | Monet   | .331        | .309         | .337                |
| FLUX  | Cézanne | .460        | .489         | .451                |
| NB2   | Monet   | .697        | .749         | .685                |
| NB2   | Cézanne | .466        | .786         | .378                |
| OAuth | Monet   | .554        | .870         | .512                |
| OAuth | Cézanne | .651        | 1.139        | .583                |

Table 3: Named/artist-free generated trace ratios, primary original-31 view. These denominators are generated controls, unlike Table 1’s original-reference denominator. Between-brief means combines within-class brief and between-class variation. Every estimate is descriptive.

Figure 4 presents this diagnostic.

### 6.3 Metric and reference choices bound robustness

All 72 named/free total-trace ratios remain below one across the fixed metric and pipeline grid. This does not mean every named collection is less dispersed than its original reference. Without texture, NB2 Monet’s named/reference ratio exceeds one in the primary and JPEG pipelines. Its primary named/free energy change is only  $-.0175$ , reversing to  $+.4159$  at 256 pixels. In contrast, both FLUX painter comparisons and NB2 Cézanne retain negative energy directions in all 12 variants per comparison. OAuth Cézanne reverses to  $+.0499$  in the primary 28-coordinate view; OAuth Monet changes direction in some 256-pixel views. No preferred view is chosen from these outcomes, and no new p-values are attached.

Primary reference texture accounts for approximately 49.0% of Monet and 47.1% of Cézanne trace, despite representing 12 of 31 coordinates. Derived-coordinate redundancy and correlations mean coordinate count is not a measure of independent stylistic evidence. Removing texture also removes substantive visual information, so its exception demonstrates measurement dependence, not invalidity of texture.

The four paid-route primary energy directions remain negative through all specified work, collection-group and content-class deletions. OAuth Cézanne crosses zero in single-work/source omissions; deleting a content class can reverse either OAuth painter comparison. All four development-painter omissions retain the paid directions, while omitting Cézanne changes its OAuth estimate to  $+.0042$ . These are sensitivities of a finite comparison, not population uncertainty intervals or independent source validation.

Figure 5 presents this diagnostic.

### 6.4 Relative painter alignment and the collection-label placebo

Primary named interactions  $I$  are  $-1.755$  for FLUX,  $-.855$  for NB2 and  $-1.802$  for OAuth. Artist-free interactions are  $-.048$ ,  $+.101$  and  $-.044$ . Named-minus-free interactions are  $-1.707$ ,  $-.956$  and  $-1.758$ , respectively. All canonical artist-free payload pairs match; separately collected outputs nevertheless have positive V-energy ( $.102$ ,  $.079$  and  $.077$ ). Stochastic sampling and service timing can produce nonzero empirical gaps under identical payloads.

The double contrast removes within-distribution terms from an uncalibrated nearest-reference comparison. It supports relative alignment for these two references, without establishing human recognition, controlling fine content, or furnishing a zero-discrepancy threshold. OAuth alignment remains conditional on its treatment-associated rendering.

Figure 6 presents this diagnostic.

### 6.5 Coverage and detection depend on the comparison design

At  $k = 3$  and full reference query counts, class-count-matched named coverage medians for Monet/Cézanne are  $.711/.703$  for FLUX,  $.447/.313$  for NB2 and  $.447/.438$  for OAuth. OAuth Monet illustrates a change of direction: all-available coverage falls from  $.553$  free to  $.474$  named, whereas class-matched medians rise from  $.421$  to  $.447$ . Coverage grows with neighborhood size. Across  $k = 3$  anchor links, 50.0% for Monet and 63.5% for Cézanne cross broad content classes. Cézanne’s three water originals force every  $k = 3$  water-anchor neighborhood to include at least one other class. A hit is not class-conditional content fidelity.

With disjoint reduced anchors,  $k = 3$  real-query coverage medians are  $.944$  for Monet and  $.933$  for Cézanne; named generated medians range  $.500-.833$  and  $.400-.800$ . At  $k = 5$ , both real medians and Cézanne FLUX named reach 1.0. This coarse neighborhood saturation does not establish equivalence. Repeated draws are finite-panel diagnostics, not independent reference replication.

Figure 7 presents this diagnostic.

Every paid output is square and every reference nonsquare, giving a square rule perfect original/generated balanced accuracy on all eight paid cells. This does not prove the feature classifier uses shape; it shows why absolute detectability cannot establish a stylistic difference. All 70 capture workflows are unresolved, and every delivery host is a Wikimedia thumbnail service. Holding institutions are not capture identities. Paid prompt arms share square geometry, strengthening their relative comparison without isolating a pure style mechanism or removing prompt-induced content differences. Constant/missing descriptors make paid-route nuisance correlations unavailable, not zero.

Across the 12 directed route/painter pairs, artist-free cross-route balanced accuracy ranges .618–.924 for linear and .620–.887 for RBF detection. Named ranges are .410–.874 and .447–.704. Naming lowers transfer accuracy in every direction for both kernels; three named linear and four named RBF cases fall below .5. Monet FLUX-to-OAuth linear transfer remains strong (.874), so this is uneven fixed-threshold transfer, not universal detector failure. Ranking summaries can move differently: Monet FLUX-to-OAuth linear mean within-fold AUC increases from .9534 free to .9676 named, despite its balanced-accuracy decline. Calibration or threshold shifts can therefore contribute; the result is not a uniform loss of discriminability. High within-route separation does not imply a calibrated, route-invariant detector. Below-chance transfer is a domain-shift result, not perceptual authenticity. Sixty of 144 case-fold instances lack an original content class; per-fold AUCs describe varying supported mixtures, whereas pooled balanced accuracy restores full finite target masses.

Figure 8 presents this diagnostic.

## 6.6 Execution sensitivity

Research attempts span 11.527 hours. Median initial randomized-group spans are 25.153 seconds for FLUX, 11.812 for NB2 and 49.950 for OAuth; maxima are 50.781, 915.340 and 953.871 seconds. Two groups cross collector boundaries; the same groups exceed 120 seconds. All eight endpoint directions remain negative under every specified timing/retry deletion. The FLUX Cézanne initial-only estimate is  $-.8938$ , versus  $-.8956$  with its selected retry. These checks preserve finite estimates but cannot test or repair unrecorded backend carryover.

## 7 Discussion and next validation target

The supported claim is that painter naming changes the tested empirical feature distributions, with improved primary proximity on both paid routes and lower aggregate spread relative to their artist-free controls. The revision explains why this should not be translated into universal style collapse: within-brief variation can increase, omitting texture changes one paid proximity direction, a named/reference spread ratio can exceed one, and fixed-threshold detector transfer loses balanced accuracy under naming. Agreement among metrics derived from the same coordinates does not supply independent validation.

FLUX has stable negative energy directions throughout this bounded grid, but a service ranking across painters, scenes or perceptual criteria is outside scope. The two-painter interaction supports relative alignment in these coordinates, while human style recognition and original-relative perceived variation remain unmeasured. Prompt content, reference selection, sampling and service rendering remain part of the observed comparison; a hidden model mechanism is not identified.

The next expansion axis is reference and construct validation. The retained metadata frame contains 112 candidate identities, all nonsquare. Forty-two are outside the panel, but include an

explicitly ineligible painting and unadjudicated candidates; they are not 42 ready additional references. Only one collection has same-content cross-painter support, one land work per painter, and this is not documented photography. A new source/content frame and declared precision target must precede an acquisition count. More generated outputs cannot fill these gaps.

The human follow-up design specifies 72 pairs of six-image named/free sets from retained outputs, preserving shared brief/repeat dependencies, with separate resemblance and nuisance judgments. Designs with 24 or 36 raters are feasibility options, not a power result. This can assess generated-only perceived variation; three Cézanne water originals cannot supply multiple disjoint six-image original boards balanced two per class. More raters would not repair missing reference replication. Stimulus-manifest execution, recruitment, consent, institutional review and judgments remain unperformed. Learned representations may provide another view later, but agreement alone cannot validate perception.

## 8 Limitations and reproducibility

The original corpus is a finite collection of available digital surrogates, not a probability sample of each painter’s oeuvre. The new reference panel is small and availability constrained. Service aliases and live model slugs lack attested immutable weight snapshots. Prompt repetition, shared briefs and service state can induce dependence. Title-based development labels and later maintainer-LLM visual annotations are not independent expert judgments. All review so far is maintainer-run LLM subagent review and coordinator checking; none is institutionally independent or a substitute for human construct validation.

The paper source, compact manifests, exact prompts, preprocessing code, report tables, split memberships and hashes are retained in the repository. Raw artwork and complete generated-image responses remain in an ignored local evidence workspace; a Git checkout alone does not reconstruct those bytes. Acquisition licenses and source identities are recorded, and failed responses remain evidence. The new study uses its own namespace and does not modify historical terminal runs.

Human evaluation is a prepared follow-up, not an executed component. Expert review could assess painter recognizability and perceived variation using blinded, balanced stimuli and independent annotation. Learned representations are also deferred; adding a learned evaluator would not automatically establish that its distance is calibrated to human style judgments. Neither a collaborator’s participation nor authorship is presumed in this prototype.

## A Revised numerical reproduction and evidence scope

The revision binds 103 inputs at source commit `f7666ae`; its freeze was committed at `c505512` before calculation. All 14 original primary cells and six named/free contrasts reproduce in the new implementation within absolute tolerance  $10^{-12}$ ; all eight all-selected timing estimates reproduce too. New report code is bound separately to the sealed JSON and its output bytes. From the repository root, replay without provider calls or image extraction:

```
uv run --locked --extra analysis python -m
latent_art_bench.painter_distribution_revision_v1.analysis check
uv run --locked --extra analysis python -m
latent_art_bench.painter_distribution_revision_v1.report_publication check
```

The JSON retains 168 metric cells, 72 prompt contrasts, 396 reference deletions, 108 specificity records, four scaler omissions, coverage memberships, nuisance associations, transfer splits/predictions and 40 timing contrasts. The protocol, validation plan and contribution



matrix are in `studies/painter_distribution_revision_v1/`. Build instructions are beside this manuscript. Every new analysis is descriptive and post-result.

Numeric replay, raw-image replay and mutable-service regeneration are different promises. Committed numeric records support the first; raw image bytes and full responses require separate access and rights review. No public raw-image release or bitwise live-service regeneration is claimed. The revision generated zero images and spent \$0. Conservative total accounting remains \$45.6819185 under the \$75 ceiling; generation was deferred for scientific reasons, not because the authorized budget was exhausted. No terminal collection should be restarted.

## B Preserved controlled-study reproduction

From the repository root, replay the completed controlled numerical analysis and all report files without provider calls:

```
uv run --locked --extra analysis python -m
latent_art_bench.painter_distribution_study_v1.analysis_publication check
uv run --locked --extra analysis python -m
latent_art_bench.painter_distribution_study_v1.main_report check
```

A separate `immediate_results generated --check` reopens retained raw bytes and recomputes all 3,018 feature vectors. The original calculation reached its JSON exporter, which rejected four NumPy Boolean rejection flags before writing any result file. The disclosed publication adapter converts only those flags to native Booleans and requires exact value equality after JSON round-trip. It uses a separate publication namespace and leaves the frozen statistical calculation unchanged. Its source, inputs and publication freeze are committed before writing the completed analysis. This representational correction is not a scientific endpoint amendment.

Historical diagnostics replay through the same package's `diagnostics check` entry point. Exact dependencies are pinned by `uv.lock`; no learned encoder is run. Current operational status and commands are in `docs/STATUS.md` and `docs/DISTRIBUTION_STUDY_WORKFLOW.md`. Raw image bytes require the separately retained local evidence workspace. No completed collector or writing stage should be restarted to reproduce a report.

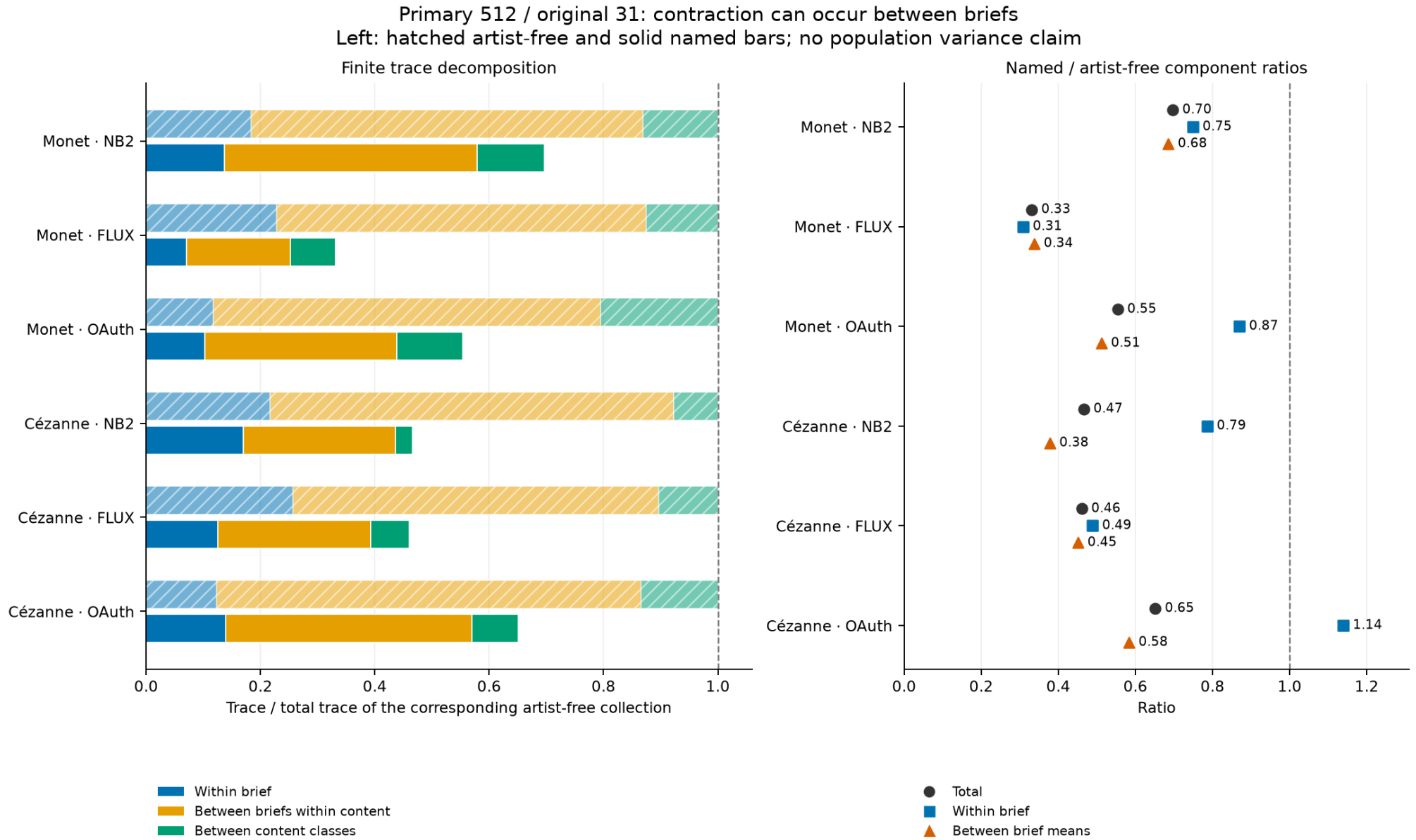


Figure 4: Finite trace decomposition under naming. Total contraction alone does not identify reduced variation at a fixed prompt.

Named minus artist-free energy across the fixed diagnostic grid  
Negative is closer; each panel uses its own metric units and color scale

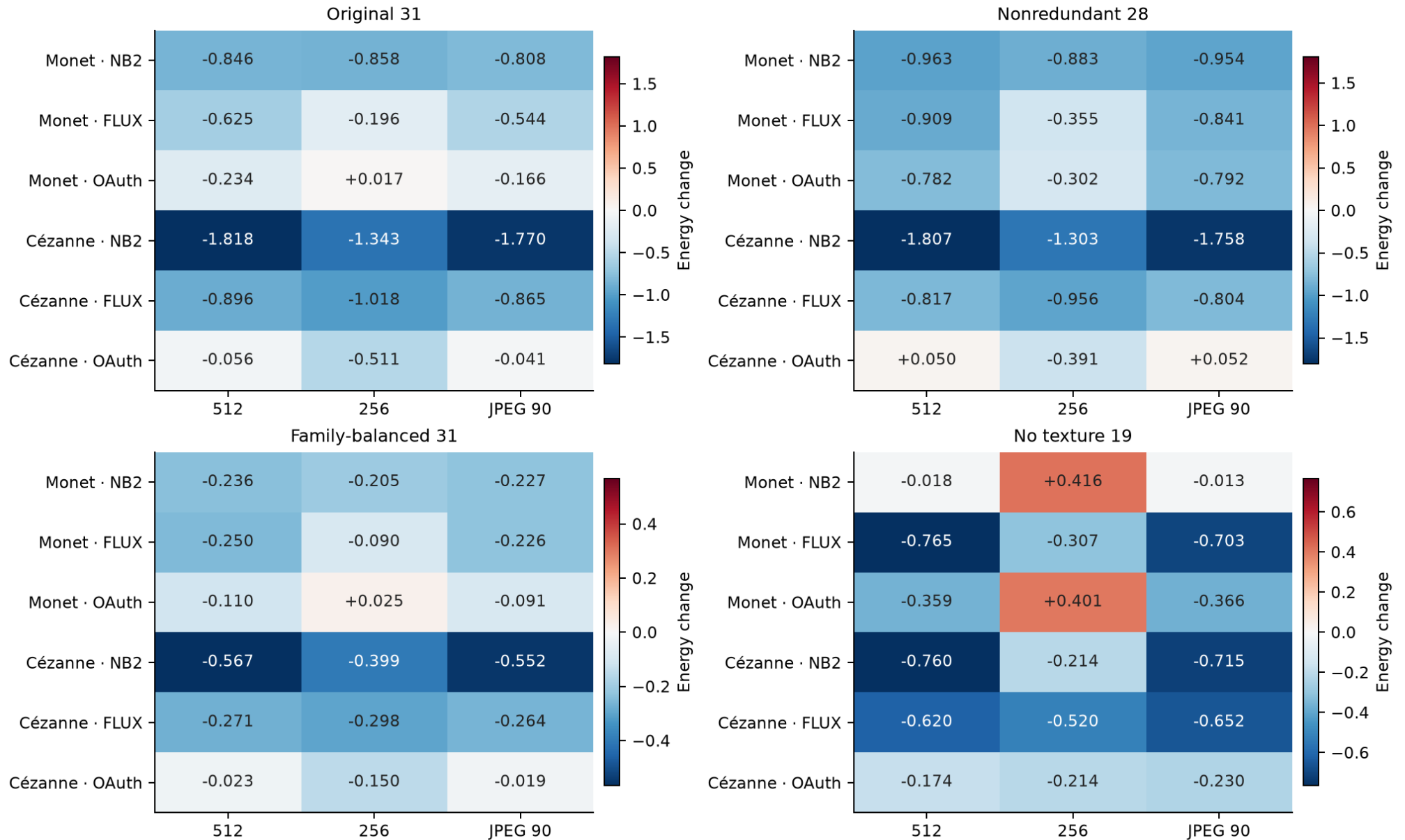


Figure 5: All fixed metric/processing views of the named/free energy contrast. Direction is compared within each metric; differently scaled energy magnitudes do not define a common effect-size scale.

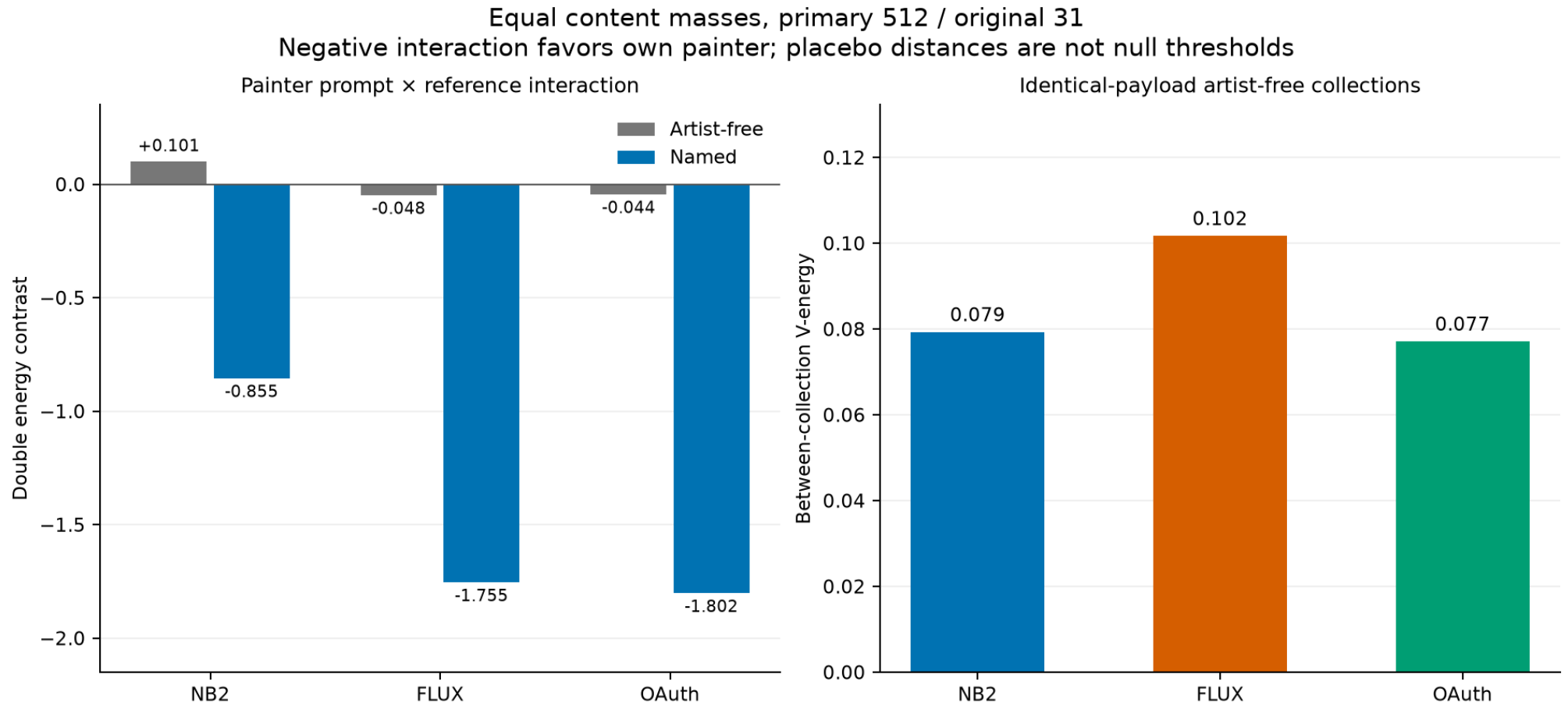


Figure 6: Painter-prompt/reference interactions and identical-payload artist-free controls. Interaction and placebo energy have different definitions and are not interchangeable thresholds.

Named collections: coverage depends on sampling, anchors and k  
 100 fixed draws; ranges are not confidence intervals; lower-row anchors differ

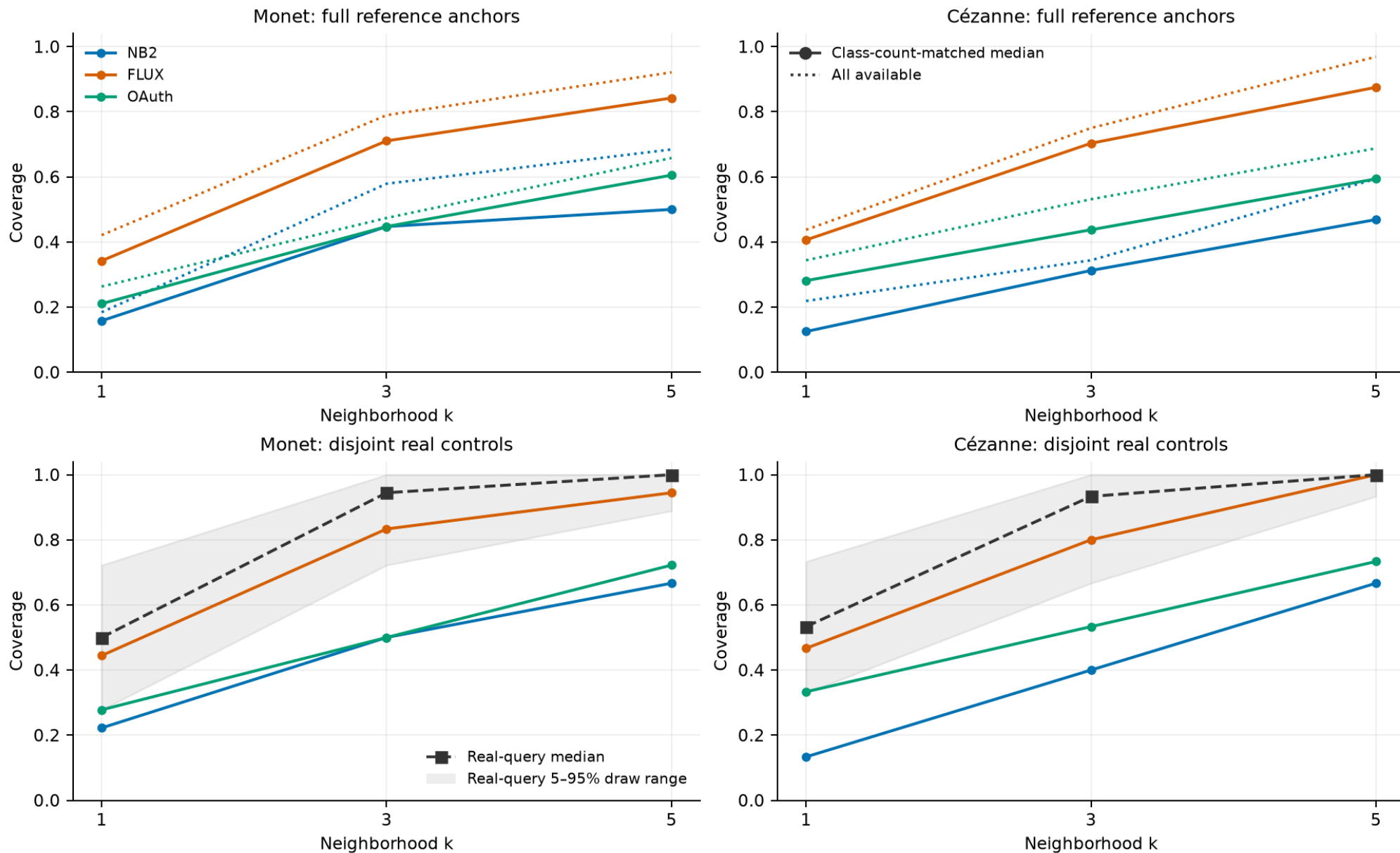


Figure 7: Coverage under fixed sample/neighborhood designs with separate disjoint real-query controls. Full-panel and reduced-anchor statistics have different reference geometry. Draw summaries are not confidence intervals.

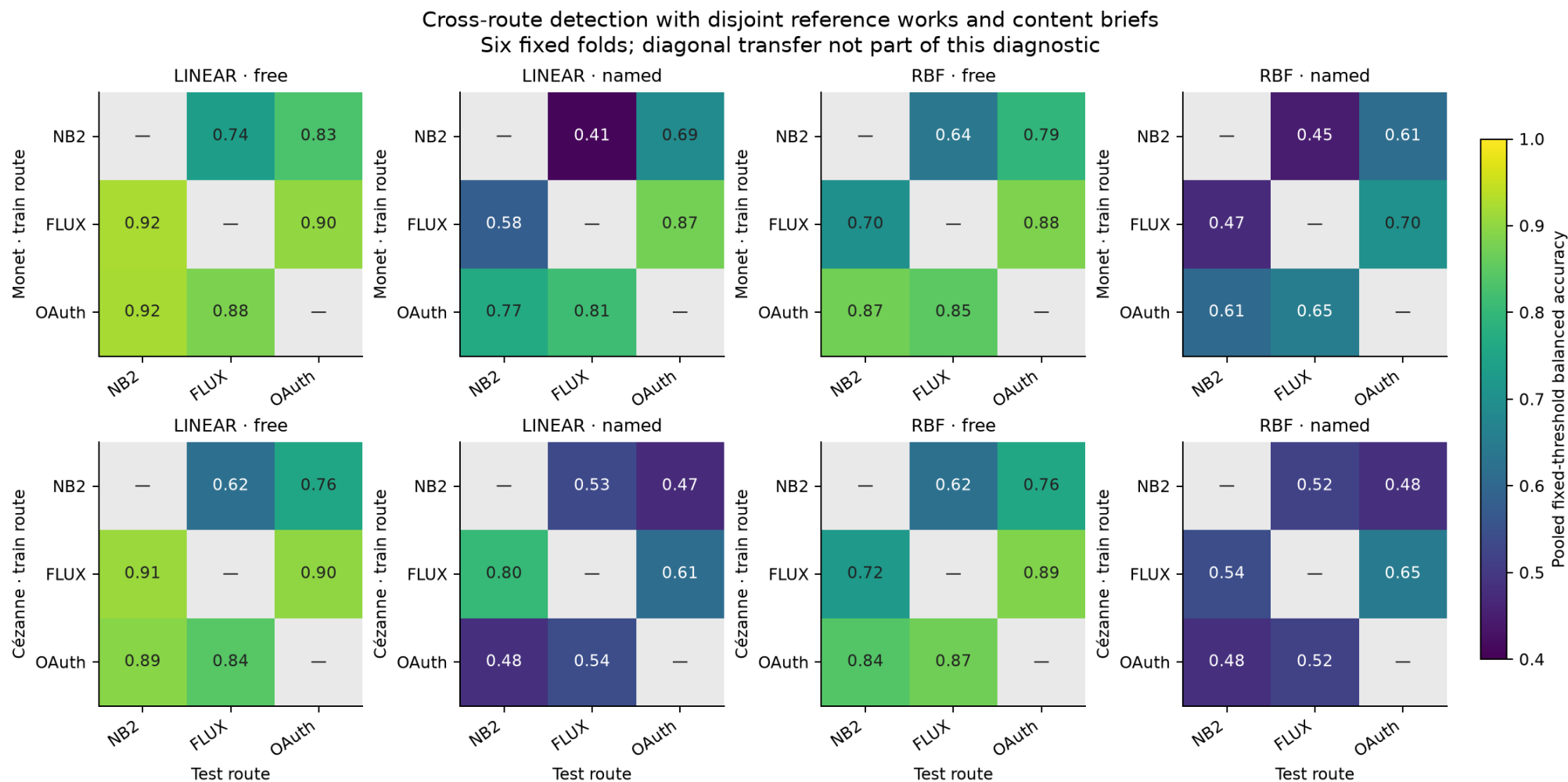


Figure 8: Directed fixed-kernel transfer, holding out original works and complete briefs. Named and free conditions differ in transferability. These are not capture-held-out evaluations.

## References

- Andrea Asperti. On the separation of human and AI-generated images in CLIP embedding space. arXiv:2608.25609, version 1, 2026. URL <https://arxiv.org/abs/2608.25609>. Preprint.
- Andrea Asperti, Franky George, Tiberio Marras, Razvan Ciprian Stricescu, and Fabio Zanotti. A critical assessment of modern generative models’ ability to replicate artistic styles. *Big Data and Cognitive Computing*, 9(9):231, 2025a. doi: 10.3390/bdcc9090231.
- Andrea Asperti, Franky George, Tiberio Marras, Razvan Ciprian Stricescu, and Fabio Zanotti. A critical assessment of modern generative models’ ability to replicate artistic styles. arXiv:2502.15856, version 1, 2025b. URL <https://arxiv.org/abs/2502.15856v1>. Inspected methods version.
- Tarian Fu, Javier Conde, Gonzalo Martínez, Pedro Reviriego, Elena Merino-Gómez, and Fernando Moral. Artificial intelligence and misinformation in art: Can vision language models judge the hand or the machine behind the canvas? arXiv:2508.01408, version 1, 2025. URL <https://arxiv.org/abs/2508.01408v1>.
- D. Kim, S.-W. Son, and H. Jeong. Large-scale quantitative analysis of painting arts. *Scientific Reports*, 4:7370, 2014. doi: 10.1038/srep07370.
- J. Kim, B. Lee, T. You, and J. Yun. Context-aware multimodal AI navigates hidden pathways in five centuries of art evolution. *Proceedings of the National Academy of Sciences*, 123(30): e2517969123, 2026. doi: 10.1073/pnas.2517969123.
- B. Lee, M. K. Seo, D. Kim, I.-S. Shin, M. Schich, H. Jeong, and S. K. Han. Dissecting landscape art history with information theory. *Proceedings of the National Academy of Sciences*, 117(43):26580–26590, 2020. doi: 10.1073/pnas.2011927117.
- Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunjey Choi, and Jaejun Yoo. Reliable fidelity and diversity metrics for generative models. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7176–7185, 2020. URL <https://proceedings.mlr.press/v119/naeem20a.html>.
- Gaurav Parmar, Richard Zhang, and Jun-Yan Zhu. On aliased resizing and surprising subtleties in GAN evaluation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. URL <https://arxiv.org/abs/2104.11222>.
- H. Y. D. Sigaki, M. Perc, and H. V. Ribeiro. History of art paintings through the lens of entropy and complexity. *Proceedings of the National Academy of Sciences*, 115(37):E8585–E8594, 2018. doi: 10.1073/pnas.1800083115.
- Gábor J. Székely and Maria L. Rizzo. Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013. doi: 10.1016/j.jspi.2013.03.018.